

Experiment 11

Question

Perform Affine Transformation on the Image.

Aim

To perform an affine transformation on an image using Python and OpenCV.

Procedure

1. Import the OpenCV (cv2) and NumPy (numpy) libraries.
2. Read the input image using `cv2.imread()`.
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Select three points from the original image.
6. Select three corresponding points in the output image.
7. Compute the affine transformation matrix.
8. Apply the affine transformation.
9. Display and save the transformed image.

Code

```
import cv2

import numpy as np

# Read image
img = cv2.imread("dog.jpg")

# Get image size
rows, cols = img.shape[:2]

# Source points
pts1 = np.float32([[50, 50], [200, 50], [50, 200]])

# Destination points
pts2 = np.float32([[10, 100], [200, 50], [100, 250]])

# Affine transformation
M = cv2.getAffineTransform(pts1, pts2)
result = cv2.warpAffine(img, M, (cols, rows))

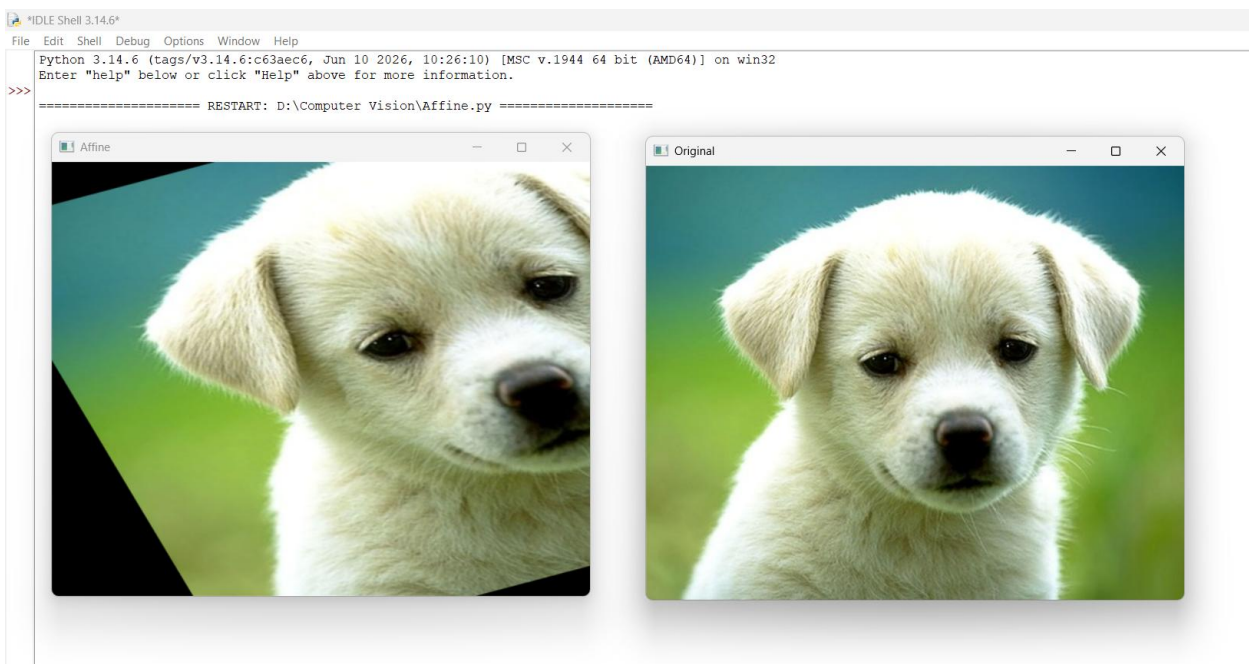
# Display images
cv2.imshow("Original", img)
cv2.imshow("Affine", result)
```

Input

Input Image: dog.jpg

Output

1. Original Image
2. Affine Transformed Image



Result

The affine transformation was successfully applied to **dog.jpg**, and the transformed image was displayed and saved.